Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/06/2025
PI Name:	LePine, Jeffrey Ph.D.		
Project Title:	Understanding and Preventing Crew Member Task Entrainment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	06/01/2015	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2023 per A. Beitman/NASA JSC (Ed., 12/1/22) NOTE: End date changed to 12/31/2022 per HRP and "in progress" information in NSSC (Ed., 3/20/2020) NOTE: Extended to 5/15/2020 per NSSC information (Ed., 6/28/19) NOTE: Extended to 5/15/2019 per NSSC information (Ed., 3/6/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:	March 2020 report: Daniel Newton, Ph.D., is now CoInvestigator on the project.		
COI Name (Institution):	Wellman, Edward Ph.D. (Arizona State University) Newton, Daniel Ph.D. (University of Iowa)		
Grant/Contract No.:	NNX15AK77G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The proposal responds to the request for research exploring task switching in crews in isolated habitats such as the International Space Station (ISS), Human Exploration Research Analog (HERA), and Nezemnyy Eksperimental'nyy Kompleks (NEK). We propose ground- and flight-based experiments to understand and mitigate the performance deficits caused as crew members switch between tasks. Drawing on our own research, as well as research conducted by other scholars, we explain how crew member entrainment is produced by deep levels of cognitive, physical, and affective engagement or immersion in tasks, which immersion makes it difficult for members to disengage from those tasks – even after they have switched to a different task. We have shown that, as a result of this immersion/engagement, entrainment causes subsequent task engagement and effectiveness to suffer, particularly when prior tasks are perceived to be incomplete. We currently investigate the potential negative effects of anticipatory entrainment in upcoming tasks on present task engagement and effectiveness. We also are in the process of building a theory of optimized task sequencing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Switching from one task to the next is a ubiquitous part of organizational life. As such, scholars have considered what makes workers effective when transitioning between different tasks (Monsell, 2003). However, a more robust understanding of the psychological connections individuals maintain with tasks—both after having performed a transition and anticipating an upcoming transition—are critical to improving and maintaining task motivation and task effectiveness. Understanding these connections could illuminate and optimize task scheduling both for crew members in orbit and individuals on earth. Monsell, S. (2003). Task switching. Trends in cognitive sciences, 7(3), 134-140. http://dx.doi.org/
Task Progress:	Research Impact / Earth Benefits: Switching from one task to another is a part of crew members' daily activity. Despite its ubiquity, a more robust understanding of entrainment, or the psychological connections crew members maintain with their tasks—both after transitioning away from a task and anticipating an upcoming task—are critical to improving motivation and effectiveness. Moreover, understanding these connections may optimize task scheduling for crew members in orbit and workers on Earth. Task Progress Aims of Proposal: Our research addresses the performance effects of entrainment as crew members engage in tasks by investing their physical, cognitive, and emotional resources in the task. Although engagement is beneficial for crew members' overall task effectiveness, it can also foster attention residue, or the inability to decouple one's energies from a prior task, which can dampen a crew member's engagement and effectiveness in subsequent tasks. We explore what can be done to mitigate these negative effects of entrainment as well as how to improve individual and team capabilities to effectively schedule and switch tasks. We specifically sought to (1) understand threats to teams during long-duration missions; (2) identify countermeasures to support team function for all phases of autonomous, long-duration missions; and (3) identify psychosocial and psychological factors that influence crew effectiveness for autonomous, long-duration missions. To probe the entrainment/task switching phenomenon, we collected 477 task transitions from 20 crew members across 5 missions in the Human Exploration Research Analog (HERA). Here we found that crew members' engagement in one task promoted their engagement in a subsequent task, as well as an indirect increase in effectiveness on that subsequent task. In other words, as crew members investigated their physical, cognitive, and emotional energy into one task, it created positive momentum that positively influenced their performance on subsequent tasks. However, task engagement also introduced attention residue that damped these positive spillover effects as crew members struggled to disconnect cognitively in switching gears between tasks. To expand on these insights, we designed a laboratory experiment in which 346 undergraduate students worked in crews of four and transitioned between various tasks. Building on the HERA findings, the laboratory experiment revealed that engaging in a task elicits positive emotions that spill over to enhance subsequent task engagement and effectiveness. However, as with the HERA crew members, task engagement also elicited attention residue, particularly when individuals did not have a sense of completion prior to switching to a subsequent task. To further deepen our understanding of these effects, we collected 83 task transitions with six astronauts aboard the International Space Station. Although a larger sample size would provide greater statistical power, the pattern of results is similar to the HERA and laboratory studies. As astronauts engaged in tasks, doing so elicited feelings of excitement and enthusiasm. Additionally, as they engaged tasks, doing so also introduced attention residue that interfered with their subsequent engagement. One crew member reported a lingering rumination after a transition. Another crew member highlighted the importance of completion. The aggregated findings from these three studies speak to the importance of several factors. First, there is a positive spillover of engagement that occurs between tasks. A primary explanation of this effect is the presence of positive emotions that ripple to enhance and maintain subsequent engagement and performance. Second, managing attention residue is essential so that the enhancing effects of task engagement can flow unobstructed throughout a workday. Third, task completion is a key solution in minimizing the negative effects of attention residue. Completion highlights the value of reducing interruptions or other forced task transitions that introduce attention residue and potential performance decrements. For more information and details on these studies and insights, please see Newton, LePine, Kim, Wellman, and Bush (2020). During conversations with crew members, several suggested that in addition to the influence of previous tasks, upcoming tasks may also influence engagement and performance. This view is understandable given that crew members engage in daily planning and frequently review the daily task schedule to facilitate task preparation and effective task accomplishment. Although research has shown the value of task planning to motivation and performance, we reasoned that looking ahead may impair performance on a current task because the crew member's attention would be attached to the upcoming task. We refer to this notion as anticipatory engagement. We examined the effects of anticipatory engagement in HERA and Nazemnyy Eksperimental'nyy Kompleks (NEK).

Task Progress:	We collected 556 task transitions from 16 crew members across 4 missions in HERA and found that anticipatory engagement in a future task impaired present task performance because it reduced crew members' engagement in the present task. We then collected 379 task transitions from 6 crew members in Scientific International Research In a Unique terrestrial Station (SIRIUS)-19 in NEK and found that anticipatory engagement only reduced crew members' engagement in the present task when the future task was complex. Although less conclusive, these findings provide some evidence of the effects of crew members' "taking their eye off the ball". Prompted by these insights, we sought to investigate the role of task planning and scheduling that might accentuate the positive effects and mitigate the negative effects of task engagement in a system of day-level tasks. The information that follows is still pre-publication and currently under review. We addressed task scheduling in terms of task meaningfulness, a core driver of engagement (Kahn, 1990). We first collected 486 day-level responses from 6 crew members in SIRIUS-21 and found that crew members were more engaged and performed at higher levels on days when their tasks decreased in sequence (i.e., the more meaningful tasks were performed earlier in the day and the less meaningful tasks were performed later in the day). Utilizing these insights from SIRIUS-21, we developed and operationalized a simple theory of the flow of engagement from one task to the next. Specifically we utilized agent based modeling to generate insights and recommendations for how crew members' tasks could be scheduled in a given day to enhance engagement and performance. Consistent with the SIRIUS-21 results, we found that day-level engagement was highest when the most meaningful tasks occurred earlier in the day and were done consecutively or close together, which notion we refer to as engagement cascade. We followed this up with a laboratory experiment in which 323 undergraduate students completed a series of four tasks that varied in terms of meaningfulness. Again, we found that overall engagement and performance were highest when the most meaningful tasks occurred earlier and were concentrated together rather than later in the day or spread out. To assess generalizability of these findings, we conducted a three-week experiment involving 132 employees who worked in multifaceted jobs. Each day, we asked participants to sequence their work tasks differently with respect to the meaningfulness of the tasks. Findings largely replicated: overall engagement and performance were higher on days when individuals sequenced tasks such that the most meaningful tasks occurred earlier and were concentrated. The implications of these findings highlight the importance of engagement cascade, in which a positive compounding of engagement is enjoyed when the most engaging tasks are concentrated and performed earlier in the day. Engagement cascade reflects a positive spillover of engagement in a task to engagement in another task (i.e., residual engagement), which over the course of a day-length sequence, results in higher day-level engagement. The heightened engagement experienced from performing meaningful tasks later in the day, or in an unconnected manner throughout the day, has less opportunity to compound and to positively impact engagement in subsequent tasks. Overall, our research sheds light on how crew members apply and maintain their psychological energies across task transitions throughout the workday. Our research highlights that although engagement in subsequent tasks may be reduced when crew members' attention remains riveted to a prior task after switching, this negative effect on engagement may be significantly reduced by features of the task set (e.g., scheduling highly engaging tasks, and tasks which can be completed earlier in the day). Our research also provides insight into how the spillover of engagement has bearing on effectiveness in multifaceted work, and by virtue of this insight, how the scheduling of tasks may be used to achieve engagement and performance bonuses. Indeed, our research indicates that beginning the day with the most engaging and meaningful work early and in a concentrated manner results in a positive spillover of engagement that is enjoyed across subsequent tasks, resulting in higher overall daily engagement and performance, relative to days structured differently. References: Kahn, W. A. (1990). Psychological conditions of personal engagement and disengagement at work. Academy of Management Journal, 33, 692-724. Newton, D. W., LePine, J. A., Kim, J., Wellman, N., & Bush, J. T. (2020). Taking engagement to task: The nature and functioning of task engagement across transitions. Journal of Applied Psychology, 105, 1-18.
Bibliography Type:	Description: (Last Updated: 11/26/2025)